

Advanced Microeconomics

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Problem Set 9: Applications of asymmetric information and Moral Hazard

1 Application

A) Consider the following Table 1, which contains data on insurance status and medical expenditures for different types of professors at Adverse Selection University (ASU). In 2018, every employee of ASU was offered a full insurance contract at no premium. In 2019, ASU charged any employee who wanted to keep health insurance a 4.000 € premium. As a result, all history professors dropped their coverage in 2019. Assume that the underlying health of ASU professors did not change much from year to year. (15 points)

	2018		2019	
	Insured	Avg. Health Expenditures	Insured	Avg. Health Expenditures
Economics Professors	Yes	5.000 €	yes	5.000 €
History Professors	Yes	3.000 €	no	2.000 €

- a.1) Is there evidence of adverse selection in this market? How do you know?
- a.2) Is there evidence of moral hazard in this market? How do you know?
- a.3) Explain how moral hazard and adverse selection combine to create a positive risk-coverage correlation in this market.
- a.4) Could the data exhibit a positive risk-coverage correlation if one of these elements were absent? How does this complicate researchers' efforts to find evidence of adverse selection?

B) Consider a Rothschild and Stiglitz model of adverse selection in which there are two types of people (robust and frail) who work for a single employer, AdvMic Inc. Both types of people are risk averse in income and have utility function of wealth:

$$U(w) = 1 - e\left(-\frac{w}{500}\right)$$

During any given year, the probability of becoming ill for the robust and frail are given by $p_{frail} = 0.2$, and $p_{robust} = 0.1$. If a person falls ill, the cost of treatment (that is, damages) is 200 €. AdvMic Inc. is deciding whether to offer their workers insurance. The firm is assumed to be risk-neutral. Each worker pays AdvMic a premium at the beginning of the year, and if the worker falls ill, the insurer

will pay the worker some amount of coverage. When they decide whether to buy insurance, workers know if they are robust or frail (though they do not know if they will fall ill during the year), but neither AdvMic Inc nor AdvMic Insurance Co can observe any worker's health status. Workers will, of course, misrepresent their type if it is in their interest to do so.

The firm decides what insurance plan (coverage and premium) to offer workers, and workers decide whether to purchase the plan or not by maximizing their expected utility. All workers have 1,000 € in initial wealth.

- b.1) AdvMic first proposes a contract for robust workers. The contract offers full coverage of the damages (i.e., an individual receives 200 € if he falls ill). What is the maximum premium AdvMic could price the contract, so that it remains attractive for robust people? Would selling this contract (at the maximum premium) to robust people be a good decision for AdvMic? You can ignore frail workers for this part of the question.
- b.2) AdvMic decides to price the Total Coverage for Robust People contract at a premium of 23 €. Would frail people want to buy this contract? If a frail person bought this contract by pretending to be robust, on average, would AdvMic make or lose money?
- b.3) To avoid this adverse selection, AdvMic Inc thinks of designing a total coverage contract for frail workers. What is the maximum premium frail workers would be willing to pay for this contract?
- b.4) If only frail workers buy the contract, what is the minimum premium AdvMic Inc could charge for a full insurance contract to just break even? Would frail people want to buy this contract? Would robust people want to buy it?
- b.5) AdvMic decides to launch two contracts: "Total Coverage" and "Minimal Coverage." The characteristics of both contracts are given in the following Table. Workers choose between total coverage, minimal coverage, and no coverage.

	Total Coverage	Minimal Coverage
Payout if ill	200 €	25 €
Premium	43 €	3 €

- b.5.1 Do frail workers prefer total coverage, minimal coverage, or no coverage? [Hint: Calculate the expected utility from each option to four decimal places.]
- b.5.2 Do robust workers prefer total coverage, minimal coverage, or no coverage?
- b.5.3 Suppose AdvMic Inc. has 100 workers, of which 75 are frail and 25 are robust. What is the firm's expected profits from offering these two contracts?